

# SPONTANEOUS DIALECT-AWARE SPEECH CORPUS FOR LOW-RESOURCE DAKHINI

A Southern Indo-Aryan Language: Methods, Challenges, and Insights

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## INTRODUCTION & KEY CONTRIBUTIONS

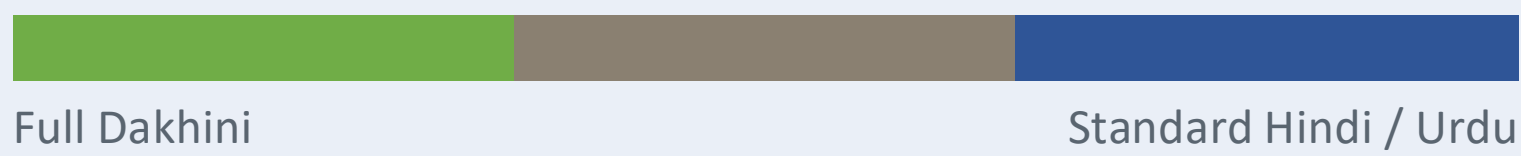
### WHAT IS DAKHINI AND WHY IS IT DIFFICULT TO COLLECT?

Dakhini (Hyderabadi Hindi) is a contact variety of the Deccan region, blending Persian, Old Urdu (Dehlavi), Kannada, Marathi, and Telugu.

Speakers exist on a continuum from full Dakhini to heavy code-switching with standard Hindi/ Urdu. Despite widespread everyday use across Southern India, it remains severely underrepresented in curated speech resources.

Our study therefore treats **linguistic authenticity** as a **first-order design constraint**, prioritizing natural conversational interaction over studio-like acoustic control.

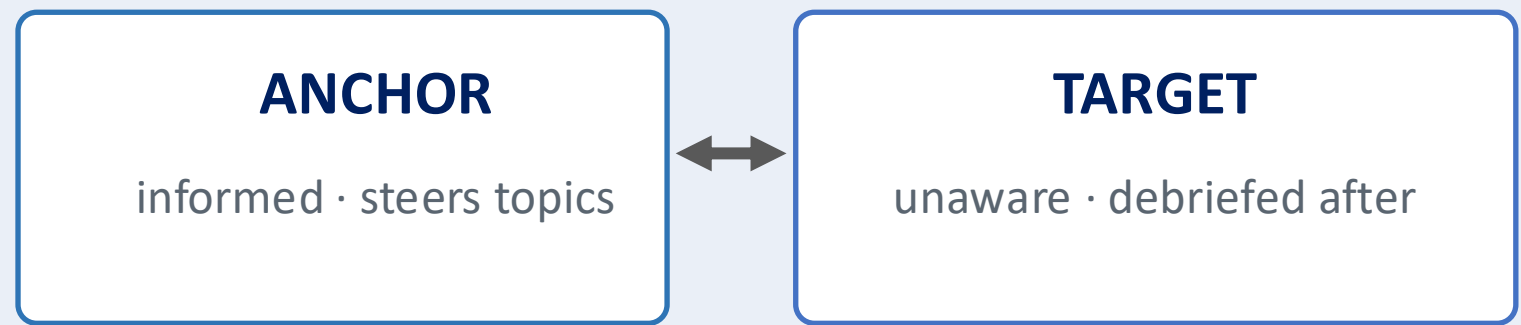
#### The Authenticity Problem



the microphone pushes speakers this way →

- Dakhini is a *non-prestige* dialect, no written standard, absent from formal education.
- Faced with a microphone, speakers drift toward the standard - the Observer's Effect - yielding data that is acoustically clean but linguistically inauthentic.

## ASYMMETRIC SPEAKER AWARENESS



- Only the anchor knows a recording is running.
- The interlocutor, i.e. the target speaker, is debriefed afterwards.
- Consent is sought before any audio is kept, and withheld consent discards the session in full.

## KEY CONTRIBUTIONS

#### First Sociolinguistic Framework

- 1 Articulates the structural & sociolinguistic challenges of collecting speech from a non-prestige, marginalised dialect community.

#### Dialect-Aware Collection Methodology

- 2 A principled, structurally-grounded pipeline for participant selection, elicitation design, and annotation rule sets.

#### Authenticity-Maximising Design

- 3 An asymmetric awareness protocol and recording design that captures genuinely unmonitored, naturalistic Dakhini speech.

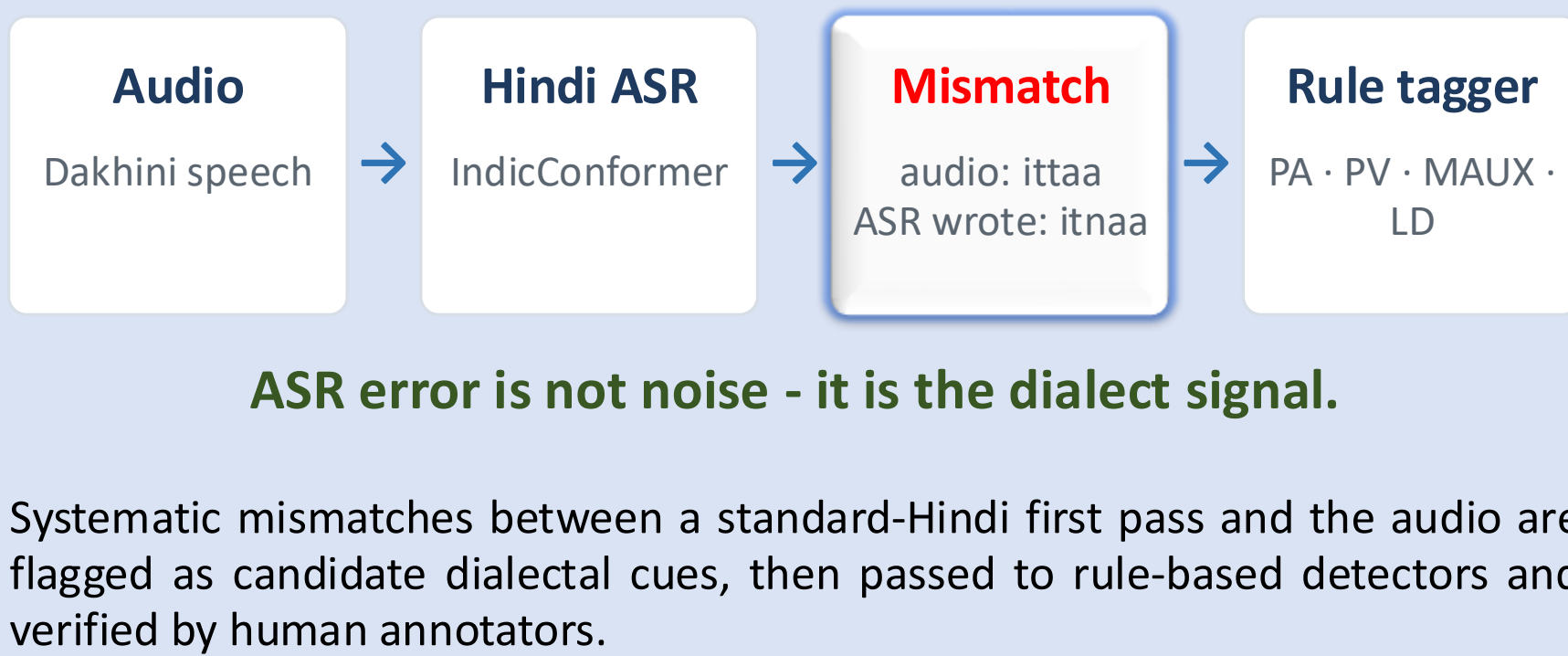
## CORPUS AT A GLANCE

|                                                  |                                                               |
|--------------------------------------------------|---------------------------------------------------------------|
| <b>160</b><br>Speakers recorded                  | <b>70/30</b><br>Male / Female split                           |
| <b>17-50</b><br>Age range (years)                | <b>20%</b><br>Higher-education background                     |
| <b>~20 min</b><br>Per session (≈2-5 min curated) | <b>4</b><br>Language fluency (Dakhini, Urdu, Telugu, English) |

## METHODOLOGY, PIPELINE & DATA

*“The collector’s linguistic competence is as important as the recording equipment.”*

## ASR AS A DIALECT DISCOVERY TOOL



### RECRUITMENT WORKFLOW

- 1 **Speaker identification**  
Hyperlocal outreach, Old City neighborhoods
- 2 **Community sessions**  
Group briefings on research intent & ethics
- 3 **Pair formation**  
*Mixed-gender, familiar speaker pairs*
- 4 **Anchor briefing**  
One speaker informed; topic menu provided

### ANNOTATION WORKFLOW

- 5 **ASR pre-transcription**  
Standard Hindi/Urdu first pass
- 6 **Rule-based tagging**  
*PA · PV · MAUX · MP-KO · MP · SF · LD*
- 7 **Human-in-the-loop**  
Prioritised review of high-tag-density segments
- 8 **Validated corpus**  
Speaker-level, annotated Dakhini dataset

## STANDARD HINDI URDU FORMS VS DAKHINI VARIANTS

| Standard Hindi/Urdu                     | Dakhini Realization | Tag   |
|-----------------------------------------|---------------------|-------|
| <b>Phonological Rules</b>               |                     |       |
| itnaa (how much)                        | ittaa               | PA    |
| us zamaane (in those days)              | uzzamaane           | PA    |
| aadmii                                  | admii               | PV    |
| <b>Morphosyntactic Rules</b>            |                     |       |
| woh gaye hai                            | woh gaye            | MAUX  |
| main khaa kar aaya                      | main khaa ko aaya   | MP-KO |
| mujh se                                 | mere se             | MP    |
| <b>Code-Switching and Lexical Layer</b> |                     |       |
| kyon (why)                              | kaiku               | LD    |
| mat karo (don't)                        | nakko karo          | LD    |
| dhire se (slowly)                       | haule               | LD    |

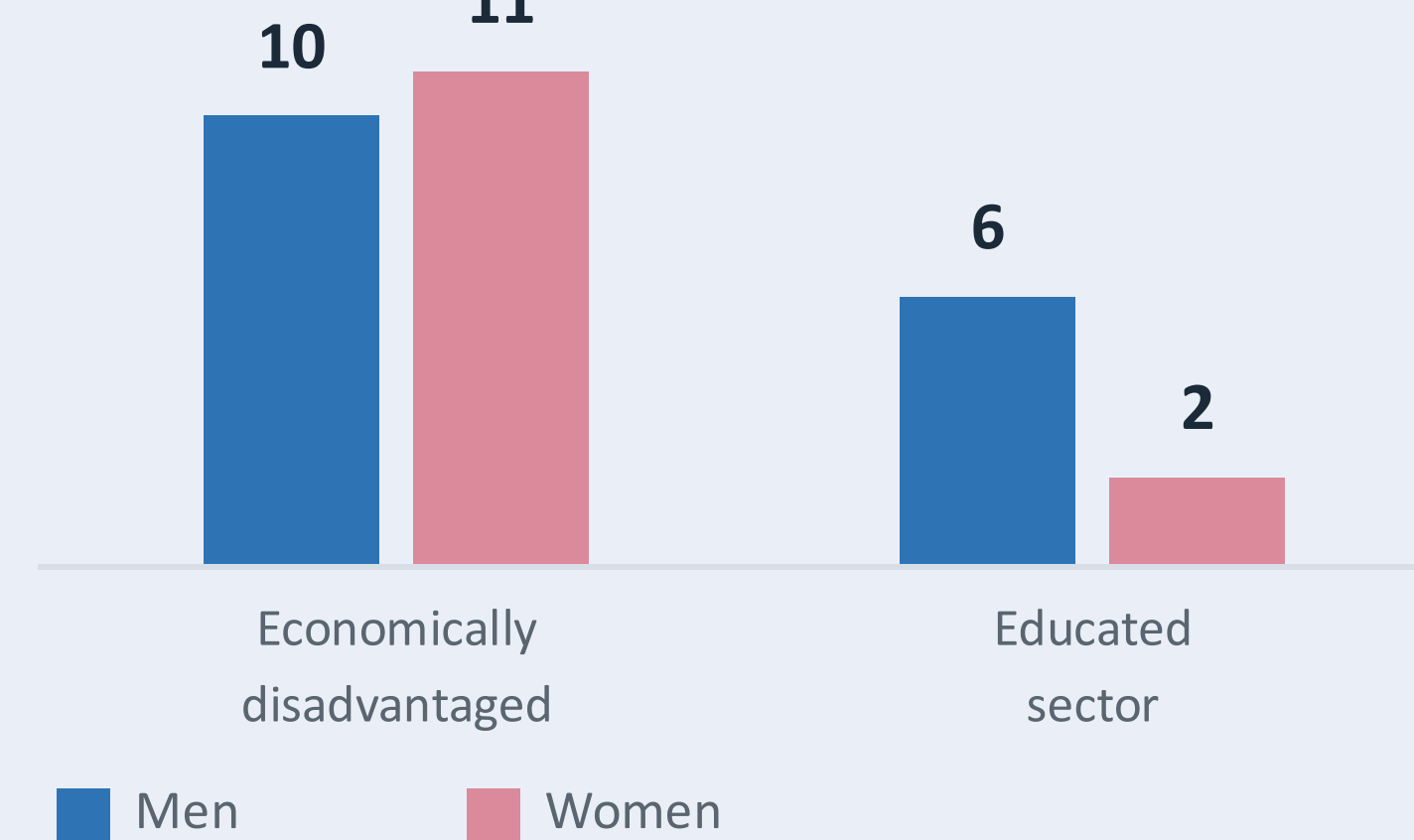
Two exemplars per rule. The full variant set and audio samples are on the demo page.

## CHALLENGES & FINDINGS

**CHALLENGES WITH RECRUITMENT** : Conventional open recruitment often failed to reach **authentic Dakhini speakers**, instead attracting participants whose speech had substantially converged toward standardized varieties. We therefore adopted **community-embedded recruitment through trusted local networks and intermediaries** to identify speakers who used Dakhini naturally across different contexts. Community information sessions were also conducted to address concerns around recording and the **stigma associated with non-prestige dialects**, while emphasizing participants’ role in documenting Dakhini as a living linguistic variety.

## SPEAKER DEMOGRAPHICS

Speakers comfortable using Dakhini in **formal** contexts (of 15)



**Educated-sector women shift away from Dakhini most sharply - 2 of 15.**

## WHY THIS WORK MATTERS

This work lays the empirical groundwork for dialect-sensitive ASR and conversational modelling in under-documented South Asian varieties and offers a replicable framework for any research team working with stigmatised, non-prestige dialect communities worldwide.

## CONCLUSION

Low-resource dialect corpus construction is not just a recording problem, it requires **linguistic and sociotechnical considerations**. Our framework combines naturalistic speaker-pair design, telephone interaction, **ASR-based dialect cues**, and linguistically informed annotation to promote **more authentic and reproducible speech resources** for under-documented varieties.

## KEY REFERENCES

- Labov, W. Sociolinguistic Patterns. UPenn Press, 1972.
- Schmidt, R. L. A Structural Analysis of Hyderabad Dakhini. PhD thesis, UPenn, 1969.
- Trudgill, P. The Social Differentiation of English in Norwich. CUP, 1974.
- Plaquet, A. & Bredin, H. Powerset multi-class cross-entropy loss for neural speaker diarization. Interspeech 2023.

## HEAR THE CORPUS



<https://mondal-anindita.github.io/Dakhini/>